Kinematics Study Notes 📝

Understanding Motion Through Graphs & Equations

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1. Kinematics Graphs Recap 📊

A quick review of fundamental graphs used in kinematics and their key properties:

- **Position-Time (s-t) Graph:** 📈
  - The **slope (ds/dt)** gives *Velocity*.
- **Velocity-Time (v-t) Graph:** 📈
  - The **slope (dv/dt)** gives *Acceleration*.
  - The **area under the curve ($\int v\,dt$)** gives *Displacement*.
- **Acceleration-Time (a-t) Graph:** 📈
  - The **slope (da/dt)** gives *Jerk*.
  - The **area under the curve ($\int a\,dt$)** gives *Change in Velocity ($v - u$)*.

2. Velocity-Displacement (v-s) Graphs 🚀

This section introduces a different type of graph and its applications.

2.1 Acceleration from v-s Graph

- When velocity (v) is a function of displacement (s), acceleration (a) is given by:
- $a = v\,(dv/ds)$ 🚀
- *Explanation:* This formula comes from the chain rule: $a = dv/dt = (dv/ds) * (ds/dt) = (dv/ds) * v$.

2.2 Problem Examples with v-s Graphs

2.2.1 Problem 1: Finding Velocity from v-s Graph

Given a linear v-s graph (line segment from (100, 50) to (200, 70)). Find velocity at s = 150.

- Observe the graph: $s = 150$ is exactly midway between $s = 100$ and $s = 200$.
- Since it's a linear graph, the velocity at $s = 150$ will be the average of velocities at $s = 100$ and $s = 200$.
- v at $s=150 = (50 + 70) / 2 = \mathbf{60}$.
- Alternatively, use the equation of a line ($y - y_1 = m(x - x_1)$) or slope equivalence.
  - Slope (m) = $(70 - 50) / (200 - 100) = 20 / 100 = 1/5$.
  - $(v - 50) / (150 - 100) = 1/5$

   - $(v - 50) / 50 = 1/5$
   - $v - 50 = 10$
   - $v = \boldsymbol{60}$.

2.2.2 Problem 2: Finding Acceleration from v-s Graph

Using the same v-s graph (linear from (100, 50) to (200, 70)), find acceleration at s = 150.

- At s = 150, $v = 60$ (from Problem 1).
- The **slope (dv/ds)** of the line segment is constant: $dv/ds = 1/5$.
- Using $a = v\,(dv/ds)$:
- $a = 60 * (1/5) = \boldsymbol{12}$.

2.2.3 Problem 3: Finding Acceleration from v-s Graph (Another Case)

Given a linear v-s graph (line segment from (0, 20) to (50, 0)). Find acceleration at s = 0.

- At s = 0, $v = 20$.
- The **slope (dv/ds)** of the line segment: $dv/ds = (0 - 20) / (50 - 0) = -20 / 50 = -2/5$.
- Using $a = v\,(dv/ds)$:
- $a = 20 * (-2/5) = \boldsymbol{-8}$.
- *Interpretation:* The negative acceleration indicates retardation or braking. 🛑

2.2.4 Problem 4: Finding Acceleration from v²-s Graph

Given a linear v^2-s graph (line segment from (0, 50) to (50, 0)). Find acceleration at s = 0.

- At s = 0, $v^2 = 50$, so $v = sqrt(50)$.
- The **slope ($d(v^2)/ds$)** of the line segment: $d(v^2)/ds = (0 - 50) / (50 - 0) = -50 / 50 = -1$.
- We know that $d(v^2)/ds = 2v\,(dv/ds) = 2a$.
- So, $2a = slope$.
- $a = 0.5 * slope = 0.5 * (-1) = \boldsymbol{-0.5}$.

2.2.5 Problem 5: Determining a-x Graph from v-x Graph

Given a linear v-x graph (decreasing from $(0, V_0)$ to $(X_0, 0)$). Identify the corresponding a-x graph.

- **Analysis of v-x graph:**
   - Velocity decreases linearly from V_0 to 0 as x increases from 0 to X_0.
   - The slope (dv/dx) is constant and negative. Let slope = $-V_0/X_0$.
   - Equation of line: $v = (-V_0/X_0)x + V_0$.
- **Deriving acceleration (a = v (dv/dx)):**
   - $dv/dx = -V_0/X_0$ (constant slope).
   - Substitute v and dv/dx into the acceleration formula:
   - $a = ((-V_0/X_0)x + V_0) * (-V_0/X_0)$
   - $a = (V_0^2/X_0^2)x - V_0^2/X_0$.
- **Interpretation of a-x equation:**
   - This is a linear equation ($y = mx + c$) for acceleration (a) as a function of displacement (x).

- The **slope (V_0^2/X_0^2)** is *positive*.
- The **y-intercept ($-V_0^2/X_0$)** is *negative*.
- Therefore, the a-x graph starts at a negative value and increases linearly with a positive slope.

2.2.6 Problem 6: Finding Acceleration from √v-s Graph

Given a linear √v-s graph with slope tan(37°) = 3/4. At some point P, √v = 2. Find acceleration at P.

- At point P: $\sqrt{v} = 2$, so $v = 4$.
- The **slope ($d(\sqrt{v})/ds$)** = 3/4.
- We need to find a relationship between slope and acceleration:
- $d(v^{1/2})/ds = (1/2) \cdot v^{-1/2} \cdot (dv/ds)$
- So, $slope = (1/2\sqrt{v}) \cdot (dv/ds)$.
- From this, $dv/ds = 2\sqrt{v} \cdot slope$.
- Now use $a = v\,(dv/ds)$:
- $a = v \cdot (2\sqrt{v} \cdot slope) = 2 \cdot v^{3/2} \cdot slope$.
- Substitute values at point P:
- $a = 2 \cdot (4)^{3/2} \cdot (3/4)$
- $a = 2 \cdot (8) \cdot (3/4)$
- $a = 16 \cdot (3/4) = \mathbf{12}$.

3. Equations of Motion

These equations describe the motion of objects under specific conditions.

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3.1 Conditions for Applicability

- The equations of motion are valid **only when acceleration (a) is constant**.
- If acceleration varies with time (e.g., $a = 2t^2$), calculus (integration/differentiation) must be used.
- The displacement (s) in these equations represents **displacement**, not always distance. Be mindful if the object changes direction.

3.2 Derivation from Velocity-Time (v-t) Graph

Consider a v-t graph where initial velocity is u, final velocity is v, time taken is t, and acceleration is constant a.

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3.2.1 First Equation: $v = u + at$

- The **slope of the v-t graph** gives acceleration.

- $a = (v - u) / t$
- Rearranging, we get: $\boldsymbol{v = u + at}$.

3.2.2 Second Equation: $\boldsymbol{s = ut + \frac{1}{2}at^2}$

- The **area under the v-t graph** gives displacement (s).
- The area can be divided into a rectangle (area = $u * t$) and a triangle (area = $\frac{1}{2} * base * height = \frac{1}{2} * t * (v - u)$).
- $s = ut + \frac{1}{2}t(v - u)$.
- Substitute $(v - u) = at$ (from the first equation).
- $s = ut + \frac{1}{2}t(at)$
- Rearranging, we get: $\boldsymbol{s = ut + \frac{1}{2}at^2}$.

3.2.3 Third Equation: $\boldsymbol{v^2 = u^2 + 2as}$

- Again, using the **area under the v-t graph** (trapezium area).
- $s = \frac{1}{2} * (\text{sum of parallel sides}) * height = \frac{1}{2} * (u + v) * t$.
- From the first equation, $t = (v - u) / a$.
- Substitute t into the area formula:
- $s = \frac{1}{2} * (u + v) * (v - u) / a$
- $s = (v^2 - u^2) / (2a)$
- Rearranging, we get: $\boldsymbol{v^2 = u^2 + 2as}$.

3.3 Problem Examples with Equations of Motion

3.3.1 Problem 1: Braking Car

A car is moving at 20 m/s. Brakes are applied, bringing it to rest in 4 seconds. Find acceleration and total distance covered.

- Given: $u = 20\ m/s$, $v = 0\ m/s$ (at rest), $t = 4\ s$.
- **Part A: Find Acceleration (a)**
  - Use $v = u + at$.
  - $0 = 20 + a * 4$
  - $4a = -20$
  - $a = \boldsymbol{-5\ m/s^2}$ (Retardation).
- **Part B: Find Total Distance Covered (s)**
  - Use $s = ut + \frac{1}{2}at^2$.
  - $s = (20)(4) + \frac{1}{2}(-5)(4)^2$
  - $s = 80 + \frac{1}{2}(-5)(16)$
  - $s = 80 - 40 = \boldsymbol{40\ meters}$.
  - Alternatively, use $v^2 = u^2 + 2as$: $0^2 = 20^2 + 2(-5)s \Rightarrow 0 = 400 - 10s \Rightarrow s = 40$ *meters*.

3.3.2 Problem 2: Accelerating Body

A body starts from rest (u=0) with a constant acceleration of 1 m/s². Find its final velocity and total distance covered after 5 seconds.

- Given: $u = 0\ m/s$, $a = 1\ m/s^2$, $t = 5\ s$.

- **Part A: Find Final Velocity (v)**
  - Use $v = u + at$.
  - $v = 0 + (1)(5) = \mathbf{5\ m/s}$.
- **Part B: Find Total Distance Covered (s)**
  - Use $s = ut + \frac{1}{2}at^2$.
  - $s = (0)(5) + \frac{1}{2}(1)(5)^2$
  - $s = 0 + \frac{1}{2}(25) = \mathbf{12.5\ meters}$.

3.3.3 Problem 3: Distance with Time Elapsed

A body is moving with a velocity of 10 m/s. It accelerates uniformly to 20 m/s over a distance of 15 m. Find the acceleration.

- Given: $u = 10\ m/s$, $v = 20\ m/s$, $s = 15\ m$.
- **Find Acceleration (a)**
  - Use $v^2 = u^2 + 2as$ (since time is not given).
  - $20^2 = 10^2 + 2(a)(15)$
  - $400 = 100 + 30a$
  - $30a = 300$
  - $a = \mathbf{10\ m/s^2}$.

3.3.4 Problem 4: Distance in the 'Nth' Second

A driver applies brakes to a car moving at 20 m/s, causing a retardation of 5 m/s². Find the distance covered in the 3rd second.

- Given: $u = 20\ m/s$, $a = -5\ m/s^2$.
- Distance in the 3rd second is $Distance(t=3s) - Distance(t=2s)$.
- Use $s = ut + \frac{1}{2}at^2$.
- **Distance in 3 seconds (s_3):**
  - $s_3 = (20)(3) + \frac{1}{2}(-5)(3)^2 = 60 - \frac{1}{2}(45) = 60 - 22.5 = 37.5\ meters$.
- **Distance in 2 seconds (s_2):**
  - $s_2 = (20)(2) + \frac{1}{2}(-5)(2)^2 = 40 - \frac{1}{2}(20) = 40 - 10 = 30\ meters$.
- **Distance in 3rd second = s_3 - s_2:**
  - $Distance = 37.5 - 30 = \mathbf{7.5\ meters}$.

3.3.5 Problem 5: Distance in 'Nth' Second (with derivation)

A body starts from rest with constant acceleration. It travels 9m in the 5th second and 25m in the 7th second. Find the acceleration and initial velocity (if not starting from rest).

- Given: $s_5 = 9m$, $s_7 = 25m$.
- Formula for distance in 'n'th second: $s\_n = u + a(n - \frac{1}{2})$.
- **For 5th second:** $9 = u + a(5 - \frac{1}{2}) \Rightarrow 9 = u + 4.5a$ (Eq 1).
- **For 7th second:** $25 = u + a(7 - \frac{1}{2}) \Rightarrow 25 = u + 6.5a$ (Eq 2).
- Subtract (Eq 1) from (Eq 2):
  - $25 - 9 = (u + 6.5a) - (u + 4.5a)$
  - $16 = 2a \Rightarrow a = \mathbf{8\ m/s^2}$.
- Substitute 'a' back into (Eq 1):
  - $9 = u + 4.5(8)$

- $9 = u + 36$
- $u = 9 - 36 = \mathbf{-27\ m/s}$. (This means the object was initially moving in the opposite direction).

3.3.6 Problem 6: Total Distance vs. Displacement (Crucial Concept) ⚠️

A body is moving with a velocity of 10 m/s and a constant acceleration of -2 m/s². Find the total distance covered in the first 6 seconds.

- Given: $u = 10\ m/s$, $a = -2\ m/s^2$.
- **Step 1: Check for change in direction.**
  - An object changes direction when its velocity becomes zero.
  - Use $v = u + at$ to find the time when $v = 0$.
  - $0 = 10 + (-2)t \Rightarrow 2t = 10 \Rightarrow t = 5$ *seconds*.
  - This means the object stops at 5 seconds and then reverses direction. The total time is 6 seconds.
- **Step 2: Calculate displacement in segments.**
  - **Displacement from t = 0 to t = 5s (s_1):**
    - $s_1 = ut + ½at^2 = (10)(5) + ½(-2)(5)^2 = 50 - 25 = 25$ *meters*.
  - **Displacement from t = 5s to t = 6s (s_2):**
    - At t=5s, $u' = 0$ (new initial velocity for this segment).
    - Time for this segment $t' = 6 - 5 = 1s$.
    - $s_2 = u't' + ½at'^2 = (0)(1) + ½(-2)(1)^2 = 0 - 1 = -1$ *meter*.
    - *Interpretation:* The object moved 1 meter in the negative direction.
- **Step 3: Calculate total distance.**
  - Total distance is the sum of the magnitudes of displacements in each segment where direction doesn't change.
  - *Total Distance* $= |s_1| + |s_2| = |25| + |-1| = 25 + 1 = \mathbf{26\ meters}$.
  - *Note:* If you just used $s = ut + ½at^2$ for 6 seconds, you would get $s = (10)(6) + ½(-2)(6)^2 = 60 - 36 = 24$ *meters*. This is the **displacement**, not the total distance.
- **Alternative Method: Using v-t Graph**
  - Plot $v = 10 - 2t$.
  - The graph is a line with y-intercept 10 and slope -2.
  - It crosses the t-axis at $t=5s$.
  - At $t=6s$, $v = 10 - 2(6) = -2\ m/s$.

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  - Area above t-axis (0 to 5s) = ½ * base * height = ½ * 5 * 10 = 25.
  - Area below t-axis (5 to 6s) = ½ * base * height = ½ * 1 * |-2| = 1.
  - Total Distance = Sum of absolute areas = 25 + 1 = **26 meters**.

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